

Homework Assignment 32

Using the Eclipse Debugger

This assignment is to give you to practice some of the basics of debugging using the Eclipse debugger. Look at the lecture “Swatting Bugs” if you need some help in using the Eclipse features. Create a new Eclipse project, **hw32**, and then download the starter file, **PascalsTriangle.java** from the exercise page and import it into your project. Add your name to the top of the file. Then, answer the following questions in your homework document. Label (number) each problem in your homework document, and supply a screenshot along with your answer when appropriate.

The program's purpose is to display Pascal's Triangle, a sequence of integers that arises in numerous areas of math and computer science, especially combinatorics and probability. When completed, the program's output should be:

Enter the number of rows (≤ 13) in Pascal's Triangle. 5

THE FIRST 5 ROWS OF PASCAL'S TRIANGLE:

```
      1
     1 1
    1 2 1
   1 3 3 1
  1 4 6 4 1
```

Entries in Pascal's triangle are indexed by integers. n is the number of the row and k is the position from the leftmost member of the row. The indexes in both directions start at zero (0), so in the fifth row listed above, $C(4,0) = 1$, $C(4,1) = 4$, and so on.

The values themselves are computed by the formula $C(n, k) = n! / (k! (n-k)!)$, which is called a *combination*. $n!$ denotes a *factorial*, that is, the product $n(n-1)(n-2)\dots(2)(1)$. The combinations can be interpreted as the number of ways to choose k elements from a collection containing n elements. When described, it is customary to say "n choose k", for instance '4 choose 3 is 4'.

If 4 objects are numbered 1 through 4, how many ways can you select 2 of them?

Question 1: Show all the possible pairings here. Then compute $C(4, 2)$ by using the formula. Does it match the number of selections you calculated manually?

As you will discover with the help of your debugger, the program to compute Pascal's triangle has some mistakes. Each provides an opportunity to learn a debugging concept in context of its use. Make sure that **PascalsTriangle.java** is selected and then:

- Select the *Run* command from the menu.
- and enter 5 for input (the number of rows for the program to display)

Question 2: What output is displayed? Show a screen-shot.

Step over / inspect

- Place a breakpoint on the **run()** method.
- Use the *Debug* command to start your debugger
- Now use the *Step Over* command several times, until the active line is
- **nrows = cin.nextInt();**
- Select *Step Over* again
- The program is now expecting keyboard input because the **nrows = cin.nextInt();** line is being executed.
- Type 5 (followed by the Enter key) for the input.

Question 3: On what line is the cursor positioned now?

- Select your compiler's *Inspect* command.
- Inspect the value of **nrows**

Question 4: What is the value for nrows shown in the debugger?

More stepping

- Select *Step Over* twice
- Look at the output for Pascal's triangle.

Question 5: Does the output have the title: TABLE 1: THE FIRST 5 ROWS OF PASCAL'S TRIANGLE ?

That is strange. The program is supposed to print three newlines, then the title. Let's find out why

- Terminate the debugging session and then restart debugging.
- Select *Step Over* until you reach the line prompting for input and then enter **5** for **nrows**
- Select *Step Over* one more time. This should skip over the line
System.out.print("\n\n\n");
- Select *Step Over* one more time.

Question 6: What line are you on now?

- Notice that you skipped over the lines:
System.out.printf("TABLE 1: THE FIRST %d", nrows);
System.out.print(" ROWS OF PASCAL'S TRIANGLE\n\n");
The reason is that two delimiters for comments had extra spaces in them.
- Change the line
/* print a title * / into
/* print a title */
and the line
/ * start a loop over the number of rows */ into
/* start a loop over the number of rows */

Question 7: When you changed the positioning of the comment delimiters, did any changes occur to the commented text? This depends on your development environment--describe what happened, if anything.

Watch

- Save your program and then select *Run* to run the program.
- Enter **5** for the number of rows. (Use **5** for input for all the runs until part 11, below).

Question 8: Does the title now show up on the screen?

Question 9: How many rows of Pascal's triangle are displayed?

- To find out why there are not five rows, let's watch a couple of variables.
- Add the Expressions pane by choosing **Window->Show View->Expressions**
- Right-click in the left-hand pane, and then select *Add watch expression*
- Add **n** to the watch list. You will likely get a message such as **n: Undefined symbol**.
The variable **n** is undefined because you haven't yet started the program.
- Select *Add watch expression* again and add **k** to the watch list.

Question 10: How many lines are now in the watch window? Shoot a screenshot.

- *Restart* and *Step Over* until you reach the line
`skip(spaces_to_skip); /* space to make a triangle */`

Question 11: What values of *n* and *k* are in the watch window?

- Keep selecting *Step Over* until you are again at the line
`skip(spaces_to_skip); /* space to make a triangle */`.

Question 12: What are the values of *n* and *k*?

- Keep selecting *Step Over* until you exit the **for** loops and arrive at the line
`System.out.println();`

Question 13: Now what are the values of *n* and *k*?

- View the program output. Notice that there are only three rows in the triangle. This is because the *n* values listed above went from 0 to 2 to 4 to 6. That is, the loop index *i* should have been increased by one each time, not two.
- Change the line of code: **for (n = 0; n < nrows - 1; n = n + 2)** appropriately.
- Select *Run* and input 5 to the program where prompted. You should now have five rows in the triangle.

Step into

- The entries in the triangle are still incorrect. To see one reason why, *Reset* and *Step Into* the program, then press *Step Over* quite a few times, until the watch window first displays:
 - **n: 2**
 - **k: 0**

You should be at line `comb = combination(n, k);`

- Press *Step Into*, to get to the line
`int combination = factorial(n) / factorial(k) * factorial(n-k);`
Notice that *Step Into* goes into each function as it is executed. Using *Step Over* always steps over the next function to the next line of code in the current procedure.
- Press *Step Into* again. You should be at the line
`int product = 1; /* accumulated product */` in the function `factorial(n)`
- Look at the variables window and find the value of **product**.
- Now press *Step Into* until you are at the line `return product;`

Question 14: What value does *n* have in the window? (Note that *k* is undefined in the watch window since *k* is not active in the function `factorial`.)

Question 15: What value does **product** have in the inspect window?

- The reason that **product** in the inspect window is incorrect is that the line

```
product = product + i;
```

should be

```
product = product * i;
```

- Make the above change in your editor and restart the program again with input of 5. Oh no! The triangle is still wrong!

Setting breakpoints

It is tedious to have to keep stepping to get inside a program. A more efficient way is to set a breakpoint at a line of interest. Then the program runs at full speed until a breakpoint is reached.

- *Reset* and make sure that **n** and **k** are still in the watch window (or add them again).
- Select the line **comb** = **combination(n,k)** ;
- Select *Add Breakpoint*.
- Select *Run* until the program stops at the breakpoint.
- Repeat selecting *Run* until the watch window displays
 - **n**: 3
 - **k**: 1
- Then *Step Into* once
- and then *Step Over* until you are at the line: **return comb**;
- Find **comb** in the variables window.

Question 16: What is the value listed for **comb**?

- The true value for the combination of 3 things taken 1 at a time is
 $3! / 1! \cdot 2! = 6 / 2 = 3$.
So, the above value of **comb** is wrong.
- Something must be wrong with our combination calculation. In fact, the programmer has omitted parentheses in the denominator for the line:
comb = **factorial(n)** / **factorial(k)** * **factorial(n-k)** ;
- Change the line to be:
comb = **factorial(n)** / (**factorial(k)** * **factorial(n-k)**) ;
- Recompile and select *Reset* and *Run*

Question 17: Is your Pascal's Triangle correct (for **nrows** = 5)?

Finally

- Select the line `comb = combination(n,k) ;`
- Select *Toggle Breakpoint*.
- Select *Run*

Question 18: Do the results look OK when you input 13 for the number of rows?
--

They should now. Shoot a screenshot showing me your code and the output. Then print your homework document as a PDF file and submit it following the instructions on the exercises page.